2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) | 979-8-3315-8618-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/EMBC58623.2025.11252937

CNN-Mamba Interaction Fusion Network for Diagnosis Retinopathy of Prematurity

Shaolong Wang, Xinyu Zhao, Zhenquan Wu, Zhen Yu, Guoming Zhang, Tianfu Wang, Baiying Lei*

***Abstract*— Retinopathy of prematurity (ROP) manifests clinically through abnormal development of the retinal capillary, ischemia, proliferative retinopathy, and retinal detachment, making it one of the leading causes of childhood blindness. Diagnosis of ROP from images of the full retinal fundus is challenging due to the presence of small peripheral lesions and underdeveloped ocular structures in premature infants. Current methods for fundus image analysis, based on Convolutional neural networks (CNNs) and transformers, often struggle to capture long-range dependencies or are limited by the complex computational demands. To address these challenges, we propose a CNN-Mamba interaction fusion network, CMIFNet, for diagnosing ROP. This network consists of a CNN-based branch for local feature extraction and a bidirectional state space model (BSSM) based branch for global information integration. We developed an interactive attention fusion module (IAFM) to enhance feature interaction and facilitate attention-based integration between the two branches. Our approach demonstrates promising recognition performance on a clinical ROP dataset.**

***keywords*— Retinopathy of Prematurity, State Space Model, Mamba, Interactive Attention**

I. INTRODUCTION

Retinopathy of prematurity (ROP) [1] is characterized by abnormal proliferation of retinal blood vessels in premature infants and low-birthweight children, making it one of the leading causes of childhood blindness [14]. Among its various forms, aggressive posterior retinopathy of prematurity (AP-ROP) [2] is the most severe. It is characterized by additional lesions in the posterior pole of the retina, neovascularization, retinal hemorrhage, and extensive avascular zones, with rapid disease progression. The prompt diagnosis and treatment of AP-ROP can lead to retinal detachment and ultimately lead to blindness. Utilizing fundus images for diagnostic screening of ROP allows for early intervention, halting disease progression. Currently, ophthalmologists rely primarily on fundus images captured by the third-generation wide-field digital pediatric retinal imaging system (RetCam) for the diagnosis of ROP. However, the International Classification of Retinopathy of Prematurity (ICROP) [3] lacks clear quantitative reference standards for AP-ROP, leading to reliance on subjective evaluations by ophthalmologists. This significantly increases the risk of missed cases and misdiagnoses. Additionally, smaller areas of ROP lesions in fundus images can be easily affected by the underdeveloped ocular structures in premature infants and artifacts from blurred images, making the screening process time-consuming and labor-intensive.

Currently, many deep learning techniques have shown promising results in the diagnosis of fundus images [15]. For instance, Hu et al. [6] introduced the max and mean aggregation operators and multi-scale information fusion methods, utilizing transfer learning to train their CNN model. Hu et al. [7] developed GLIM-Net, which utilizes time positional encoding and time-sensitive self-attention for glaucoma prediction from irregularly sampled fundus images. Lei et al. [8] designed a dual-branch network that integrates features from both ViT and CNN branches. However, existing fundus image analysis methods are often limited by the small receptive field of convolutional operations, making it challenging to capture long-range dependencies. Although Transformer architectures can mitigate some of these limitations, their complex computational requirements remain a significant constraint. Even with CNN-Transformer fusion architectures, the issue of high computational overhead persists. Therefore, there is an urgent need for a novel method to address this challenge.

Recently, the Mamba model [9], a hardware-efficient state space model (SSM), has garnered substantial research interest due to its ability to capture long-range dependencies while maintaining linear time complexity. [4], [13] , making it a new research hotspot. For instance, ViM [10], inspired by ViT [19], utilizes location embedding to integrate spatial information and employs a two-way SSM to process non-causal image sequences. VMamba [21] expands image blocks into sequences along both horizontal and vertical dimensions, scanning in both directions. EM-NET [5] designs channel squeeze-reinforce Mamba blocks to integrate and select channels, leveraging frequency domain information for multi-scale feature learning in segmentation. Inspired by Mamba, we propose a novel interactive fusion network, CMIFNet, for the diagnosis of ROP. The network comprises two branches: a CNN branch for detailed feature extraction

This work was supported in part by the National Natural Science Foundation of China (Nos. 82271103, 82301269, 82301226, 82401315), Sanming Project of Medicine in Shenzhen (No. SZSM202311018), Guangdong Basic and Applied Basic Research Foundation (No. 2022A1515012326), Shenzhen Medical Research Fund (Nos. C2301005, A2403020), Shenzhen Key Medical Discipline Construction Fund (No. SZXK038), and Shenzhen Fund for Guangdong Provincial High-level Clinical Key Specialties (No. SZGSP014).

Shaolong Wang, Tianfu Wang, Baiying Lei are with the National-Regional Key Technology Engineering Laboratory for Medical Ultrasound, Guangdong Key Laboratory for Biomedical Measurements and Ultrasound Imaging, School of Biomedical Engineering, Health Science Center, Shenzhen University, Shenzhen, China, 518060.

Xinyu Zhao, Zhenquan Wu, Zhen Yu, Guoming Zhang are with Shenzhen Eye Hospital, Jinan University, Shenzhen Eye Institute, Shenzhen, China, 518040.

*Correspondence should be addressed to Baiying Lei (leiby@szu.edu.cn)

979-8-3315-8618-8/25/$31.00 ©2025 IEEE
Authorized licensed use limited to: SRM University Amaravathi. Downloaded on September 06,2026 at 15:00:26 UTC from IEEE Xplore. Restrictions apply.

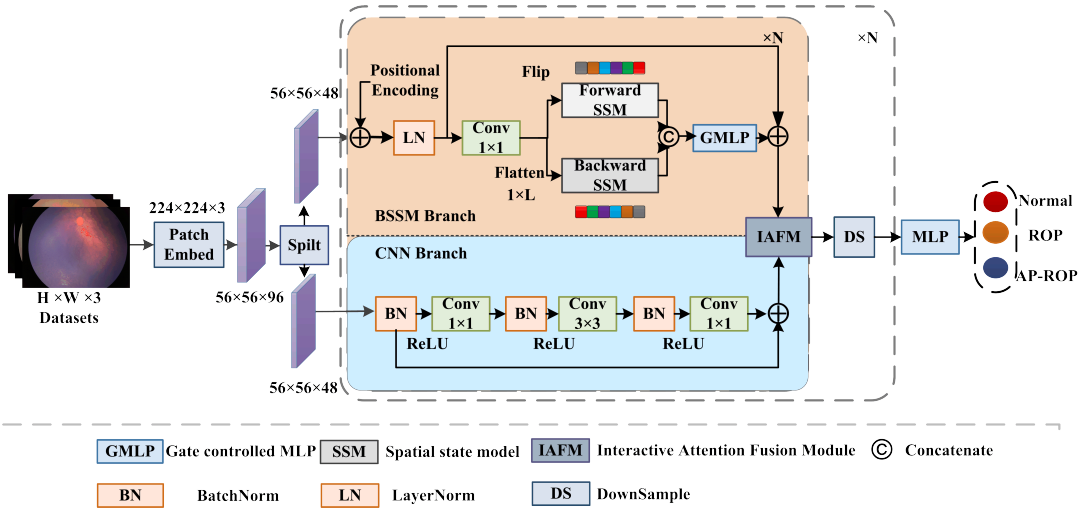


Fig. 1. The overview of the proposed CMIFNet for diagnosis of ROP

and an BSSM branch for capturing global features from fundus images. To better aggregate the extracted features, we designed an interactive attention fusion module (IAFM) to facilitate feature interaction and refine the effective features from both branches. Finally, the effectiveness of our method was validated using clinical ROP datasets.The main contributions of this work are as follows:

1) We propose a ROP diagnostic network, CMIFNet, which integrates features from CNN and BSSM, fully leveraging both local and global features of fundus images.

2) We designed a novel IAFM to learn the feature information between the two branches, enhancing feature selection effectiveness and maximizing the utilization of spatial features in fundus images.

3) Experimental results on clinical ROP datasets demonstrate that our method outperforms other methods.

II. METHODOLOGY

A. The framework of CMIFNet

Fig.1 shows the detailed structure of CMIFNet. The model comprises a BSSM branch, a CNN branch, and an IAFM. The BSSM branch consists of two SSM pathways that extract global information from the fundus image from two directions, including the overall structure and background information of the retina. Meanwhile, the CNN branch focuses on local features, such as morphological changes in new blood vessels and retinal hemorrhages at specific locations. The IAFM module is designed to integrate both global and local information. Finally, a multilayer perceptron (MLP) is used as the classifier head to obtain the final classification results.

B. BSSM and CNN module

The SSM model was originally designed to process 1-D dense data in natural language processing (NLP) and is unable to capture the spatial features of images. To address this limitation, we followed the design principles of ViM [10] and adopted the SSM bidirectional scanning method, where image patches are compressed into 1-D data. The sequences of the images are flipped to perform both forward and backward SSM scans, allowing for the perception of spatial contextual features. This approach naturally captures long-range dependencies while preserving the integrity of the information. To better integrate the features extracted from both directions, we introduced a gMLP [11] after the BSSM. This design employs MLPs and gating mechanisms to extract spatial context from the images. Additionally, before inputting the image features into the BSSM, we add positional encoding to the 1-D image tokens, enabling the model to learn spatial sequence information. This improves its capacity to capture directional cues and effectively utilize the contextual features of the images.

For the CNN branch, we processed the divided patches using resBlocks from ResNet50 to extract local information between the patch blocks. Compared to applying convolutions over the entire image, this approach reduces the computational load, lowers the dependence on channel complexity, and enables the extraction of more advanced semantic features.

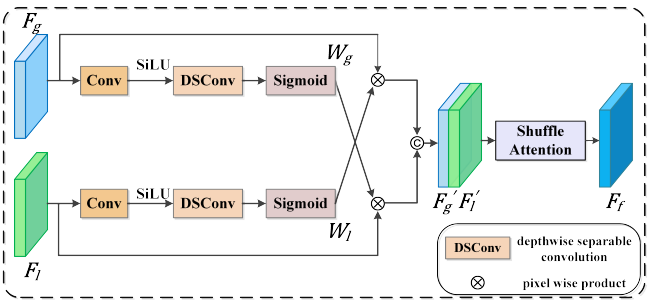


Fig. 2. Illustration of the IAFM

C. Interactive Attention Fusion Module

Simple concatenation is insufficient for effectively fusing the features captured by the BSSM blocks and CNN blocks.

Authorized licensed use limited to: SRM University Amaravathi. Downloaded on September 06,2026 at 15:00:26 UTC from IEEE Xplore. Restrictions apply.

TABLE I

PERFORMANCE COMPARISON BETWEEN OUR PROPOSED METHOD AND OTHER METHODS ON ROP DATASETS.

Methods	AP-ROP vs Normal				AP-ROP vs ROP				ROP vs Normal			
	ACC	Pre	F1	AUC	ACC	Pre	F1	AUC	ACC	Pre	F1	AUC
ResNet-50 [17]	90.97 ±2.82	93.08 ±2.69	94.18 ±1.81	95.10 ±2.62	88.45 ±2.97	91.52 ±2.52	91.81 ±2.17	94.56 ±2.26	79.33 ±4.81	73.85 ±7.19	72.68 ±7.12	87.26 ±5.24
ConvNeXt-T [18]	91.61 ±3.69	88.73 ±5.46	86.28 ±6.63	95.40 ±4.08	87.33 ±3.78	90.11 ±2.95	91.10 ±2.66	93.37 ±3.31	77.86 ±5.36	70.68 ±6.59	73.78 ±1.44	85.34 ±6.79
ViT [19]	88.73 ±2.55	91.27 ±1.07	92.03 ±3.35	94.48 ±2.47	87.62 ±2.89	90.63 ±2.49	91.25 ±2.09	93.67 ±2.56	76.75 ±4.55	71.56 ±6.70	72.29 ±3.97	83.92 ±5.83
ViM [10]	91.23 ±3.28	87.72 ±4.85	85.85 ±5.78	95.20 ±3.27	88.06 ±3.14	90.18 ±2.51	91.65 ±2.22	94.15 ±2.63	78.92 ±5.50	72.39 ±6.89	72.38 ±7.87	86.37 ±6.23
Vmamba [21]	92.70 ±3.82	**94.09 ±3.68**	**95.51 ±2.27**	96.10 ±4.02	87.74 ±3.75	90.20 ±3.72	91.46 ±2.56	93.89 ±3.29	82.67 ±4.96	75.46 ±3.37	**75.82 ±2.75**	88.94 ±3.57
CMIFNet	**95.09 ±2.54**	93.18 ±2.68	93.08 ±2.41	**96.69 ±5.43**	**89.19 ±1.47**	**93.63 ±2.66**	**92.05 ±2.79**	**96.42 ±0.91**	**85.14 ±2.85**	**76.76 ±3.18**	74.64 ±3.96	**91.54 ±4.15**

To enhance feature fusion between the two branches, we designed an IAFM for feature interaction and integration. Specifically, we first input the local features F_l and global features F_g extracted from the CNN and BSSM branches into a depthwise separable convolution, followed by an activation function to obtain the corresponding weights W_g and W_l. This computational process can be expressed as follows:

$$W_g = \delta(DSConv(SiLU(Conv(F_g)))) \quad (1)$$

$$W_l = \delta(DSConv(SiLU(Conv(F_l)))) \quad (2)$$

Where the SiLU $(\cdot)$ is the SiLU activation function, δ is the Sigmoid function. Next, following the simple gate idea of NAFNet [12], a simple gating mechanism is implemented through matrix dot products, mapping the screening of global and local features to the corresponding branches, and finally applying the global and local feature maps to their respective branches using the channel connection and shuffle attention [16] is employed to complete the aggregation of the feature maps. This process can be described as follows:

$$F_f = SA(Cat((W_g \otimes F_l), (W_l \otimes F_g))) \quad (3)$$

Here, F_f denotes the final output of the IAFM, where $Cat(\cdot)$ denotes the concatenation of the gated-weighted features F_l' and F_g', and $SA(\cdot)$ represents the shuffle attention mechanism. The fused features are downsampled and subsequently passed into the next CNN and BSSM modules to reduce model complexity.

III. EXPERIMENTS AND RESULTS

A. Datasets

The fundus image data used in this study were collected using a RetCam3 device at a specialized ophthalmology hospital and annotated by experienced ophthalmologists. A total of 6,499 retinal images without ROP, 1,698 images at the AP-ROP stage, and 4,033 images at the common ROP stage were utilized. Image acquisition was approved by the institutional ethics committee and conducted in accordance with the Declaration of Helsinki. Based on this dataset, we conducted binary classification experiments across three category pairs: AP-ROP vs. Normal, AP-ROP vs. ROP, and ROP vs. Normal.

B. Experimental setup

We employed data augmentation techniques including random cropping, horizontal flipping, and random rotation. To conserve computational resources, we resized the input images to 224×224. The learning rate was set to 0.0001, and the batch size was set to 32. We used the Adam optimizer with first- and second-order momentum decay values of 0.9 and 0.999, respectively. Binary cross-entropy was employed as the loss function. The experiments were conducted on two NVIDIA TITAN XP GPUs, each with 12 GB of memory. We employed a 5-fold cross-validation method for comparison, which helps mitigate overfitting in the model. In this study, we used accuracy (Acc), precision (Pre), F1-score (F1, the harmonic mean of precision and recall), and the Receiver Operating Characteristic – Area Under the Curve (ROC-AUC) as evaluation metrics to assess and compare the model's performance.

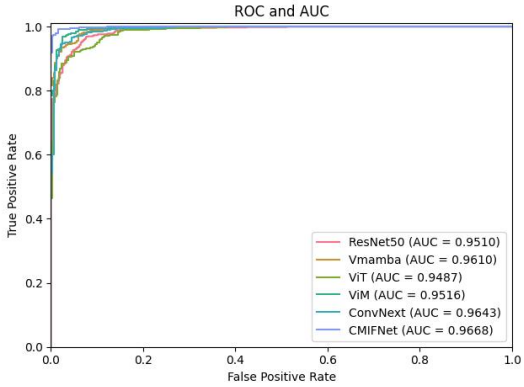


Fig. 3. ROC curves of different classification methods.

TABLE II

ABLATION ANALYSIS OF DIFFERENT MODULES ON AP-NORMAL

BSSM	IAFM	ACC	Pre	F1	AUC
		90.28	91.56	90.66	92.25
✓		93.96	92.37	91.26	93.75
✓	✓	95.09	93.18	93.08	96.69

Authorized licensed use limited to: SRM University Amaravathi. Downloaded on September 06,2026 at 15:00:26 UTC from IEEE Xplore. Restrictions apply.

C. Results

As shown in Table 1, the proposed CMIFNet demonstrates outstanding performance across three classification tasks. In the AP-ROP vs Normal task, CMIFNet achieves an accuracy of 95.09%, significantly surpassing other baseline models. This superior performance can be attributed to CMIFNet's dual-branch architecture: the CNN branch captures localized pathological features, such as neovascular morphology, while the BSSM branch integrates global retinal structural information through long-range dependency modeling. From a model category perspective, Mamba-based models (ViM [10], VMamba [21]) outperform traditional CNNs (ResNet-50 [17], ConvNext [18]) in global feature modeling. However, their accuracy in the AP-ROP vs. Normal task remains lower than that of CMIFNet. This indicates that relying solely on global features may overlook localized pathological details, whereas CMIFNet compensates for this limitation through its dual-branch interaction mechanism. Fig. 3 illustrates the ROC curves of the evaluated models. CMIFNet achieves the highest Area Under the Curve (AUC) value of 0.9688 among all models. This exceptional AUC performance highlights CMIFNet's ability to effectively integrate both localized pathological features and global retinal structural patterns, which are crucial for the accurate classification of subtle ROP manifestations.

D. Ablation study

To validate the effectiveness of various modules in CMIFNet, we conducted ablation experiments on the AP-ROP vs Normal task to analyze their individual contributions. As shown in Table 2, the bidirectional SSM with a gating mechanism improved the model's accuracy by 3.68% compared to the unmodified SMM module. Additionally, the diagnostic accuracy increased by 1.13% after incorporating the IAFM module. This indicates that IAFM effectively facilitates interaction and fusion between the dual-branch features, selectively integrating missing features from each branch while suppressing irrelevant ones. Overall, the addition of these two modules enhanced the model's classification performance, demonstrating their contribution to improving feature extraction.

IV. Conclusion

This paper introduces a novel CMIFNet for the diagnosis of ROP, which effectively combines the strengths of CNNs and BSSMs. By integrating localized features from fundus images with global spatial information, CMIFNet addresses the limitations of existing methods in capturing both fine-grained details and long-range dependencies. Specifically, we propose an IAFM, which dynamically retrieves complementary information from the CNN and BSSM branches, enabling the consolidation of multi-scale features. Extensive experiments on a clinical ROP dataset demonstrate that CMIFNet achieves state-of-the-art performance in ROP classification tasks.

References

[1] A. Hellström, L. E. Smith, and O. Dammann, "Retinopathy of prematurity," *The lancet*, vol. 382, no. 9902, pp. 1445–1457, 2013.

[2] Y. Ahn, K. Hong, H. Yum, J. Lee, K. Kim, Y. Youn, and S. Park, "Characteristic clinical features associated with aggressive posterior retinopathy of prematurity," *Eye*, vol. 31, no. 6, pp. 924–930, 2017.

[3] I. C. for the Classification of Retinopathy of Prematurity *et al.*, "The international classification of retinopathy of prematurity revisited," *Archives of Ophthalmology (Chicago, Ill.: 1960)*, vol. 123, no. 7, pp. 991–999, 2005.

[4] D. Han, Z. Wang, Z. Xia, Y. Han, Y. Pu, C. Ge, J. Song, S. Song, B. Zheng, and G. Huang, "Demystify mamba in vision: A linear attention perspective," *arXiv preprint arXiv:2405.16605*, 2024.

[5] A. Chang, J. Zeng, R. Huang, and D. Ni, "Em-net: Efficient channel and frequency learning with mamba for 3d medical image segmentation," in *International Conference on Medical Image Computing and Computer-Assisted Intervention*. Springer, 2024, pp. 266–275.

[6] J. Hu, Y. Chen, J. Zhong, R. Ju, and Z. Yi, "Automated analysis for retinopathy of prematurity by deep neural networks," *IEEE transactions on medical imaging*, vol. 38, no. 1, pp. 269–279, 2018.

[7] X. Hu, L.-X. Zhang, L. Gao, W. Dai, X. Han, Y.-K. Lai, and Y. Chen, "Glim-net: chronic glaucoma forecast transformer for irregularly sampled sequential fundus images," *IEEE Transactions on Medical Imaging*, vol. 42, no. 6, pp. 1875–1884, 2023.

[8] H. Lei, J. Zhao, H. Xie, Y. Liu, G. Zhang, and B. Lei, "Dual-branch feature interaction network with structure information learning for retinopathy of prematurity classification," in *2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*. IEEE, 2023, pp. 1230–1235.

[9] T. Dao and A. Gu, "Transformers are ssms: Generalized models and efficient algorithms through structured state space duality," 2024. [Online]. Available: https://arxiv.org/abs/2405.21060

[10] L. Zhu, B. Liao, Q. Zhang, X. Wang, W. Liu, and X. Wang, "Vision mamba: Efficient visual representation learning with bidirectional state space model," *arXiv preprint arXiv:2401.09417*, 2024.

[11] H. Liu, Z. Dai, D. So, and Q. V. Le, "Pay attention to mlps," *Advances in neural information processing systems*, vol. 34, pp. 9204–9215, 2021.

[12] L. Chen, X. Chu, X. Zhang, and J. Sun, "Simple baselines for image restoration," in *European conference on computer vision*. Springer, 2022, pp. 17–33.

[13] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," in *The International Conference on Learning Representations (ICLR)*, 2022.

[14] J.-A. Quinn, F. M. Munoz, B. Gonik, L. Frau, C. Cutland, T. Mallett-Moore, A. Kissou, F. Wittke, M. Das, T. Nunes *et al.*, "Preterm birth: Case definition & guidelines for data collection, analysis, and presentation of immunisation safety data," *Vaccine*, vol. 34, no. 49, pp. 6047–6056, 2016.

[15] S. Li, H. Singh, and A. Grover, "Mamba-nd: Selective state space modeling for multi-dimensional data," *arXiv preprint arXiv:2402.05892*, 2024.

[16] Q.-L. Zhang and Y.-B. Yang, "Sa-net: Shuffle attention for deep convolutional neural networks," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 2235–2239.

[17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[18] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2022, pp. 11 976–11 986.

[19] D. Alexey, "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint arXiv: 2010.11929*, 2020.

[20] Z. Tu, H. Talebi, H. Zhang, F. Yang, P. Milanfar, A. Bovik, and Y. Li, "Maxvit: Multi-axis vision transformer," in *European conference on computer vision*. Springer, 2022, pp. 459–479.

[21] Y. Liu, Y. Tian, Y. Zhao, H. Yu, L. Xie, Y. Wang, Q. Ye, and Y. Liu, "Vmamba: Visual state space model," 2024. [Online]. Available: https://arxiv.org/abs/2401.10166

Authorized licensed use limited to: SRM University Amaravathi. Downloaded on September 06,2026 at 15:00:26 UTC from IEEE Xplore. Restrictions apply.